Design of Strong-Arm Latch Comparator for Low Power 12-bit SAR ADC

1st Pinninti Kishore
Dept. of ECE
VNR Vignana Jyothi Institute of Engineering and Technology
Hyderabad, India
kishore_p@vnrvjiet.in

2nd Vishlavath Tarun
Dept. of ECE
VNR Vignana Jyothi Institute of Engineering and Technology
Hyderabad, India
tarunvislavath18@gmail.com

3rd Segi Reddy Venkata Satya Siva
Dept. of ECE
VNR Vignana Jyothi Institute of Engineering and Technology
Hyderabad, India
satyasivasegireddy1201@gmail.com

4th Sreemayi Kompella
Dept. of ECE
VNR Vignana Jyothi Institute of Engineering and Technology
Hyderabad, India
sreekompe@gmail.com

2024 International Conference on Integrated Circuits and Communication Systems (ICICACS) | 979-8-3503-1755-8/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICICACS60521.2024.10498940

***Abstract*—As wireless communication and the need for low powered circuits is evolving, there is a need for an efficient design for ADCs. One of the crucial components of an ADC is a comparator and its primary purpose is to compare two voltage levels and generate a logical output based on their relationship. Dynamic comparators have a significant advantage of being more power efficient than ordinary comparators. In this paper, Strong-Arm latch comparator circuit is designed. The Strong-Arm latch comparator is designed in 32 nm CMOS technology in the Synopsys tool. This architecture for a comparator is a crucial electronic circuit designed for high speed and precise comparison of two input signals. Its functional components include an input stage that receives signals for comparison, a Strong-Arm stage responsible for swiftly determining the larger input, a latch stage employing positive feedback to capture and maintain the comparison result and an output stage indicating the outcome. The strong-arm phase actively drives one input to hasten the comparison process, enhancing the response time. The Strong-Arm latch has proven to have the least power dissipation. The proposed design has 1.16 µW of power dissipation when $V_{in_P} > V_{in_n}$ and 1.21 µW of power dissipation when $V_{in_p} < V_{in_n}$ in the evaluation phase. The proposed design has low power dissipation of 251 nW, and it has a time delay of 0.02 nsec.**

Keywords*—Comparator, latch, Analog-to-digital converter, successive approximation register, low-power.

I. INTRODUCTION

As innovations in wireless communication are advancing, the requirements for ADCs increase as they deal with higher data speeds and therefore comparator performance should be more efficient too. A comparator is an electronic circuit built with operational amplifiers (Op-Amps) that compares two input signals and generates an output. In another paper titled "Low-power comparator design for SAR ADC" [3], a low-power comparator is presented, specifically designed for a Successive Approximation Register (SAR) ADC. This design is particularly pertinent for devices relying on non-rechargeable batteries, such as medical implants. The resulting output signal from the comparator indicates whether one input is greater or lesser than the other. A non-inverting comparator is configured when an input voltage is connected to the non-inverting terminal of an operational amplifier (op-amp), and a reference voltage is applied to the inverting terminal. In this setup, the op-amp is termed non-inverting because the input voltage for comparison is directed to the non-inverting terminal. The circuit diagram for a non-inverting comparator is illustrated in Fig. 1. In this report, the efficiency of a comparator's performance is evaluated with a focus on speed, accuracy, and power consumption. The key parameters under consideration for this comparator analysis are power consumption and delay.

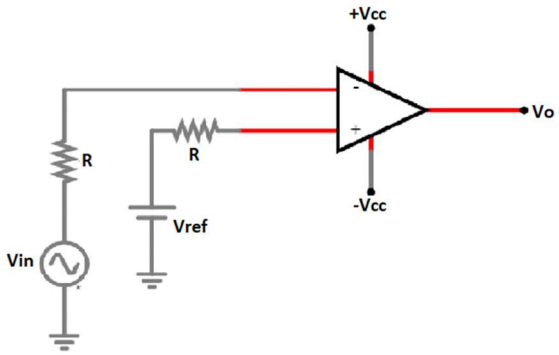


Fig. 1. Comparator

A non-inverting comparator functions in a relatively straightforward manner. Depending on the values of the input voltage V_{in} and the reference voltage V_{ref}, a non-inverting comparator produces one of the two output values: either $+V_{sat}$ (positive saturation voltage) or $-V_{sat}$ (negative saturation voltage) as shown in Fig. 2.

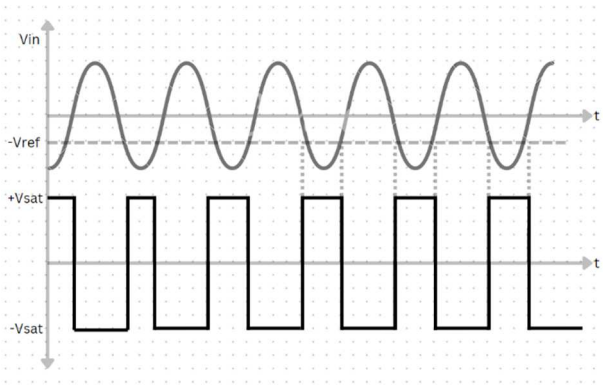


Fig. 2. Comparator waveform

Dynamic comparators represent a unique type of comparator that optimizes power consumption by integrating the circuit's clock. In dynamic comparators, the clock is utilized to trigger a reset on one edge and an assessment on the other. The comparator employs two outputs, with one mirroring the logic output and the other presenting its inverse,

979-8-3503-1755-8/24/$31.00©2024 IEEE

Authorized licensed use limited to: Tamkang Univ.. Downloaded on June 24,2026 at 04:34:00 UTC from IEEE Xplore. Restrictions apply.

as illustrated. The dynamic latched comparator operates through two distinct phases: the evaluation phase and the reset phase. During the reset phase, the comparator endeavours to reset all internal voltage levels, while in the evaluation phase, it compares the inputs and produces the corresponding logic output based on the larger input voltage. The ideal characteristics of the comparator input and output are illustrated in Fig. 3 If the reset phase is not sufficiently long, the voltage levels can act as memory and influence subsequent comparisons.

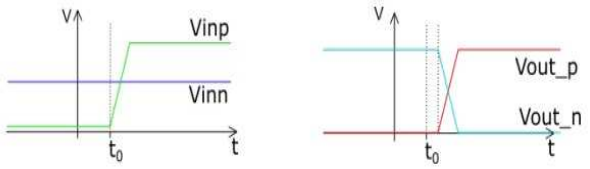


Fig. 3. Dynamic comparator output

The Strong-Arm latch is considered one of the fundamental dynamic comparators. It stands as a crucial electronic circuit in analog signal processing, specifically designed for the rapid and precise comparison of two input signals. Its architecture encompasses distinct stages, initiating with an input stage that receives the signals earmarked for comparison. Following this, the strong-arm stage takes charge, actively determining the larger input and optimizing response time. The subsequent latch stage utilizes positive feedback to capture and sustain the comparison result until the next cycle. The final communication of the comparison outcome is facilitated by the output stage. Optionally, hysteresis integration enhances stability, particularly in scenarios where input signals closely align. Crafted to meet the demands of applications requiring rapid and precise signal comparisons, the Strong-Arm latch comparator holds significant relevance in analog-to-digital converters and clock and data recovery circuits. Its notable features include a small input referred offset, high input impedance, rail-to-rail output, and a lack of static power consumption. These distinctive attributes contribute to its widespread adoption as a robust latch, comparator, or sense amplifier. As a result, the Strong-Arm latch is commonly integrated into Flip-Flop and analog-to-digital converter (ADC) circuits. The persistent need for compact, speedy, and energy-efficient electronics remains crucial, especially in the context of the growing interest in wearable devices, the Internet of Things (IoT), and low-power applications.

II. Existing Comparator Designs

The paper titled "Comparator design for high-speed ADCs" [1] conducts a comparative analysis of various comparators, evaluating their operational performance and power dissipation. On a related note, the paper titled "Design of low-power and high-speed comparator for A/D applications" [2] introduces an innovative CMOS comparator design featuring a clock-driven preamplifier-latch circuit tailored for Sigma-delta ADC integration. The design achieves a noteworthy reduction in power consumption (102 µW) and an increase in speed (125 MHz), all while maintaining high accuracy. Simulations conducted in a 0.18 µm CMOS Technology using Cadence demonstrate improvements over previous methodologies, highlighting the efficacy of the proposed approach. In another paper titled "Low-power comparator design for SAR ADC" [3], a low-power comparator is presented, specifically designed for a Successive Approximation Register (SAR) ADC. This design is particularly pertinent for devices relying on non-rechargeable batteries, such as medical implants. Implemented in 180 nm technology, the comparator attains high speed (29 MHz), minimal power consumption (1.8 µW), and a low offset voltage of 5mV. Its simple topology, featuring 8 MOS devices in the sub-threshold region, underscores its efficiency for low-power applications. The paper titled "A Low-power High-speed Comparator for Analog to Digital Converters" [4] introduces a two-stage dynamic comparator designed for optimal balance between low power consumption and high speed. This design incorporates PMOS transistors and controlled delays. Simulations conducted in a 0.18 µm CMOS environment reveal a substantial 50% reduction in power consumption and a faster comparison time of 210 ps compared to conventional designs. This innovation provides flexibility in adjusting offset voltage and power consumption according to specific speed requirements. In another paper titled "Low-power self-calibrating dynamic comparator for high-speed ADCs" [5], a dynamic latched comparator with self-calibration is presented, achieving a low offset voltage of 1.69 mV at 1 sigma. This is in stark contrast to the 13.7 mV offset voltage observed without calibration. The comparator is designed for high-speed ADCs and is characterized by low noise and efficient power consumption. The self-calibration feature enhances precision and contributes to the overall performance improvement. In the paper titled "A comparator with reduced delay time in 65-nm CMOS technology for supply voltage down to 0.65 V" [6], a comparator is designed with a modified latch specifically for low-supply-voltage operation. This design has been fabricated in low-power 65 nm CMOS and characterized using an on-chip delay-time measurement circuit. The emphasis is on achieving reduced delay time, especially for supply voltages as low as 0.65 V. The article discussing comparators of latch type [7] focuses on the design of high-speed and low static power dissipation comparators. Different types of comparator circuits are explored, considering the trade-off between power consumption and propagation latency. Various designs, such as the Latch type voltage sense comparator, Dynamic two-stage energy-efficient latched comparator, and modified two-stage latched comparator with inverters, are analyzed for their performance parameters. Notably, the power consumption of the Latch type voltage sense comparator is reduced by transitioning to the Dynamic two-stage energy-efficient latched comparator, albeit with an increase in delay. The Modified two-stage latched comparator with inverters is another design aimed at achieving lower power consumption at the expense of increased latency. The research aims to enhance the performance and decrease power consumption in comparators, crucial components in digital circuits. Two comparators were meticulously designed through pass transistor logic (PTL) in both static and dynamic configurations, employing CMOS and CNTFET technologies. A comprehensive evaluation, utilizing the Synopsys HSPICE tool and 32 nm technology, revealed that comparators one and two outperformed their counterparts when considering transistor count, power consumption, and latency. This emphasizes the significance of considering these factors during the construction of digital circuits [8]. The paper addresses the power consumption bottleneck in high-resolution Successive Approximation Register (SAR) analog-to-digital converters (ADCs) intended for low-power applications. The proposed solution involves optimizing the

Authorized licensed use limited to: Tamkang Univ.. Downloaded on June 24,2026 at 04:34:00 UTC from IEEE Xplore. Restrictions apply.

distribution of comparators unevenly across different bit steps. This optimization strategy results in substantial power savings, reaching up to 50% and 60% for 10-bit and 12-bit SAR ADCs with 10 and 12 comparators, respectively. Additionally, the study suggests that achieving near-ideal performance may be possible with as few as two comparators, offering a potential reduction in implementation complexity [12]. The paper titled "Low power, low noise edge race comparator for SAR ADCs" introduces an innovative voltage comparator known as the edge-race comparator (ERC). This comparator distinguishes differential input voltages by generating and measuring the distance between two propagating edges in inverter loops. The ERC dynamically adjusts its noise, power consumption, and delay based on input voltage, making it well-suited for low-power, high-resolution SAR ADCs. In comparison to the edge-pursuit comparator (EPC), it exhibits a 3.39 times faster speed at a 1-mV input and reduces energy consumption by 2.73 times at the same input voltage, owing to a distinctive configuration. Experimental results in a 40-nm CMOS process demonstrate a 7.5 times reduction in energy at the least significant bit (LSB) and 3.85 times increase in ADC sampling rate [13]. The article titled "A low power Dynamic Comparator for a 12-Bit Pipelined Successive Approximation Register (SAR) ADC" introduces a 210 MS/s 12-bit pipelined SAR ADC tailored for mobile applications. It identifies the dynamic comparator as the primary contributor to high power consumption in the circuit. To address this issue, the proposed design incorporates a double-tail dynamic comparator with an output buffer, effectively reducing the circuit's power consumption. The maximum power consumption of the comparator in use has been measured at 402.3 pW [14]. In the paper titled "A low-noise high-speed comparator for a 12-bit 200-MSps SAR ADC in a 28-nm CMOS process," the authors present a high-speed, low-noise comparator utilizing 28-nm bulk CMOS technology and functioning with a 0.9-V supply voltage. This comparator is specifically tailored for incorporation into a 12-bit, 200-MSps successive-approximation-register (SAR) analog-to-digital converter (ADC). Simulation results reveal an input-referred noise of 163 μV and a reset-out delay of 110 ps for a 100 μV input differential voltage. The energy consumption per conversion is reported at 595 fJ/conv, and the Figure-of-Merit (FOM) is computed as 15.8 nJ·μV², demonstrating superior performance compared to existing state-of-the-art solutions [15].

III. Proposed Design

The strong-arm latch comparator is a pivotal electronic circuit widely utilized in analog signal processing, particularly for high-speed and precise comparisons between two input signals. The latch topology of Strong-Arm as shown in Fig. 4 discovers widespread use as a sensory enhancer, a comparator, or just a sturdy latch with extreme sensitivity. The phrase "Strong-Arm" honors the application of this circuit in electronic devices. Comprising distinct stages, this comparator begins with an input stage receiving signals to be compared, followed by a strong-arm stage that actively determines which input is larger, optimizing response time. The latch stage employs positive feedback to capture and sustain the comparison result until the next cycle. An output stage then communicates the final comparison outcome. Optionally, hysteresis can be integrated for stability in scenarios where the input signals are closely matched. This comparator's design is tailored for applications demanding rapid and accurate signal comparisons, making it instrumental in various fields, including analog-to-digital converters and clock and data recovery circuits.

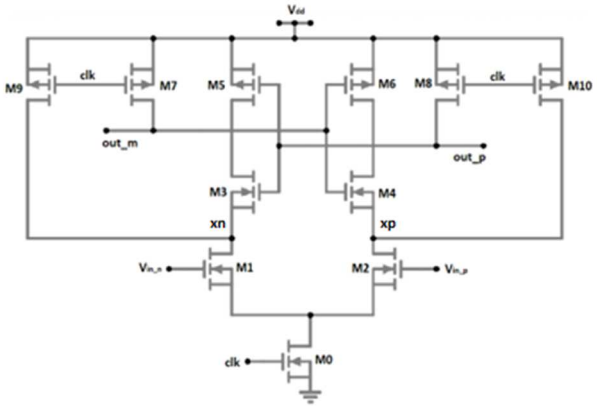


Fig. 4. Strong-Arm Latch Comparator

The clock signal in dynamic comparators serves a dual purpose: it initiates the evaluation process on one clock edge and the reset process on the other. In this configuration, two outputs are obtained, with one representing the logic output and the second providing the inverse of the first. During the initial phase when the clock signal (CLK) transitions low, nodes xn, xp, out_m, and out_p are precharged to V_{dd}. Subsequently, as CLK goes high, transistors M1 and M2 start discharging nodes xn and xp until they approach $V_{dd} - V_{th}$. Activation of M5 or M6, triggered by the switching on of M3 and M4, pushes one output back to V_{dd} while the other output is discharged, leading to a logical result. A notable advantage of this dynamic comparator design is its independence from static electricity, as the M0 tail transistor is toggled on and off by the clock signal rather than being biased to a constant current. This approach utilizes a clock input signal, eliminating the need for a static power supply. The Strong-Arm latch comparator exhibits several notable advantages in its design. One of its primary strengths lies in its high-speed operation, making it well-suited for applications that demand rapid response times. The architecture of latch-based comparators, including the Strong-Arm variant, often allows for effective implementation of hysteresis, contributing to improved noise immunity and enhanced stability in dynamic environments. Furthermore, the simplicity of the latch-based design contributes to ease of implementation, potentially reducing manufacturing costs. Additionally, Strong-Arm latch comparators can be optimized for low power consumption, rendering them suitable for energy-efficient devices and battery-operated applications. These advantages collectively position the Strong-Arm latch comparator as a compelling choice in scenarios where speed, power efficiency, and simplicity are critical design considerations. As is the case in nearly every electronic design, achieving low power consumption is a primary objective. One notable advantage of the dynamic latched comparator, in contrast to pre-amplifier-based comparators, is its ability to operate without consuming static power, assuming leakage currents are negligible [12]. The power consumption of this comparator is typically calculated by taking various factors and considerations into account [9-11]. Equation (1) shows the calculation of power consumption.

$$P_{avg} = I_{Vdd,avg} \times V_{dd} \quad (1)$$

Authorized licensed use limited to: Tamkang Univ.. Downloaded on June 24,2026 at 04:34:00 UTC from IEEE Xplore. Restrictions apply.

The more intriguing parameter in this context may be the energy per cycle, which is typically calculated by considering the energy consumed by the dynamic latched comparator in each operational cycle. The calculation in (2) involves accounting for factors such as the dynamic power dissipation during the evaluation and reset phases, as well as any other relevant energy-consuming elements within the comparator circuit. The formula for energy per cycle can be derived by multiplying the power consumption by the duration of each cycle.

$$E_{cmp} = \frac{I_{Vdd,avg} \times V_{dd}}{F_{clk}} \quad (2)$$

To have a fair comparison between the circuits, the power consumption from the clock path is also included in (3) and calculated by the following formula where I_{clk} is the current of the clock input.

$$E_{clk} = \int \left| \frac{I_{clk} \times V_{dd}}{2 \times N_{cycles}} \right| dt \quad (3)$$

The resulting total energy per cycle is calculated by.

$$E_{cycle} = E_{cmp} + E_{clk} \quad (4)$$

IV. Results and Analysis

The figures below depict the simulation results of the Strong-Arm Latch Comparator. The hysteresis of the comparator is characterized when the input voltages are set at 6% V2n (in0) and 60% V2n (in0). The determination of hysteresis involves sweeping the input in a triangular wave with small steps. By examining the plotted outputs in Fig. 3, specific input values are identified where the comparator transitions from 0 to 1 and vice versa. The hysteresis is then precisely calculated as the difference between these input values. This approach provides a comprehensive understanding of the comparator's behavior under varying input conditions. In a dynamic latched comparator, two distinct phases govern its operation: the reset phase and the evaluation phase. The reset phase initiates when the CLK signal transitions low, causing the outputs out_m and out_p to go high as they are charged to V_{dd}. During this phase, the comparator is prepared for the upcoming comparison. The actual comparison of the inputs takes place in the evaluation phase, which commences when the CLK signal transitions high. In this phase, the output out_m goes high when $V_{in_p} > V_{in_n}$, indicating that the positive input voltage is greater than the negative input voltage. Conversely, the output out_P goes high when $V_{in_P} < V_{in_n}$, signifying that the positive input voltage is less than the negative input voltage. These output states reflect the result of the comparison process during the evaluation phase. Table 1 shows the power dissipation in the pre-charge and evaluation phase.

TABLE I. Power Dissipation of Proposed Circuit

Power Dissipation		
Pre-Charge	Evaluation	
	$V_{in_p} > V_{in_n}$	$V_{in_p} < V_{in_n}$
251.056 nW	High:1.16 µW	High:1.16 µW
	Low: 1.21 µW	Low: 1.21 µW

The above table represents the power dissipation of the proposed architecture of the strong-arm latch comparator in 32-nm technology. The power dissipation of the circuit in pre-charge is 251.056 nW, the power dissipation in the evaluation phase is 1.16 µW and observed time delay is 10ns. The figures 5,6 and 7 show the transitions in the output according to the changes observed in the input voltages of V_{in_p} and $V_{in_n}$and the CLK.

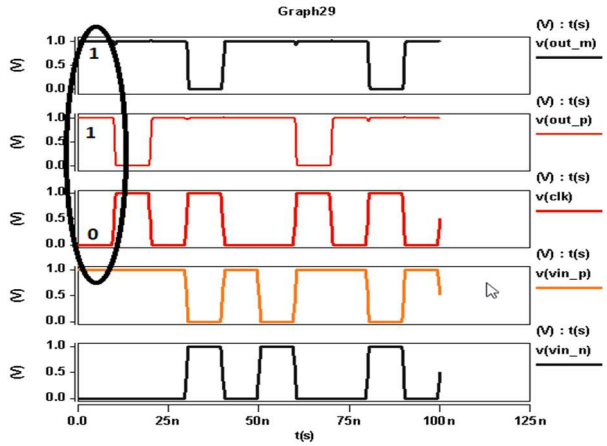


Fig. 5. Strong-Arm Latch Comparator precharge phase

When the CLK is set to low or is at a logic level zero, transistors M0, M7, M8, M9 and M10 are turned off and charge is accumulated at both the output nodes out_m and out_p and return a logic 1. This denotes the precharge phase.

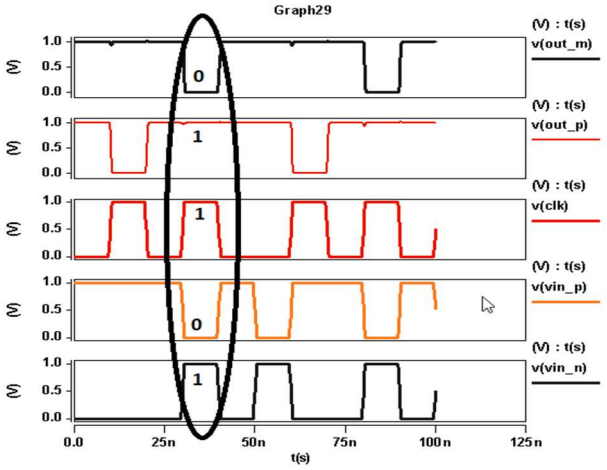


Fig. 6. Strong-Arm Latch Comparator evaluation phase 1

The evaluation phase starts when CLK turns 1. When V_{in_n} is greater than V_{in_p} , charge in the out_m is depleted into the ground and returns logic 0. The transistor M2 stays turned off and has high charge in out_p thus returning logic 1.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on June 24,2026 at 04:34:00 UTC from IEEE Xplore. Restrictions apply.

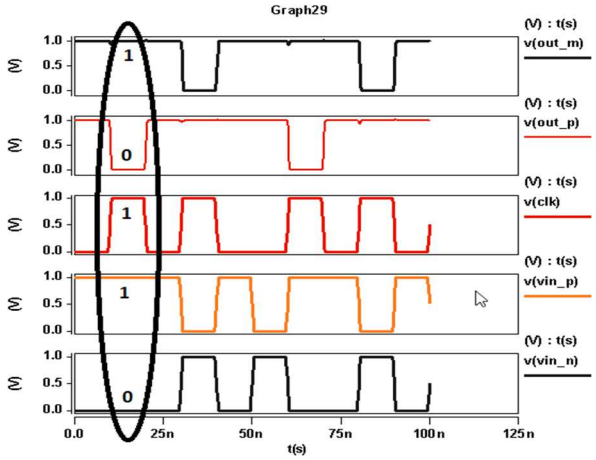


Fig. 7. Strong-Arm Latch Comparator evaluation phase 1

Another case in evaluation phase is when V_{in_n} is less than V_{in_p}. The transistor M1 is turned off making out_m produce logic 1 and the charge at out_p flows to the ground to return logic 0 at the output.

TABLE II. COMPARISON OF POWER DISSIPATION WITH OTHER COMPARATORS

Power Dissipation		
	$V_{in_p} > V_{in_n}$ (µW)	$V_{in_p} < V_{in_n}$ (µW)
Strong-arm latch comparator	1.16	1.21
Double Tail Comparator	2.13	2.21
Triple Tail Comparator-I	2.20	2.28
Triple Tail Comparator-II	4.29	4.46

The above table represents the comparison of power dissipation of strong -arm latch comparator with three other different comparators. Strong – Arm Latch comparator has less power dissipation when compared with the double tail comparator, triple tail comparator-I and triple tail comparator-II. Therefore, it is deduced that the suggested comparator stands out as the optimal choice for low-power consumption applications. Consequently, we have decided to employ this comparator in the construction of a low-power Successive Approximation Register (SAR) Analog-to-Digital Converter (ADC). Table 3 shows the comparison of various parameters with the existing models.

TABLE III. COMPARISON OF FEW PARAMETERS WITH OTHER COMPARATORS

S.no	Title	Power Dissipation (µW)	Delay (nano secs)	Technology Used (nm)	Supply Voltage (V)
1	Design of Low Power and High-speed CMOS comparator	102	8	180	1.8
2	Low Power comparator for SAR ADC	90	4	180	1
3	Design of low power and High-speed comparator with sub -32 nm double gate mosfet	8.91	0.0351	32	1
4	A high speed and low power 32nm FinFET dynamic Latch Comparator Design	45.64	0.0075	32	0.8
5	Analysis and Design of a Low-Voltage Low-Power Double-Tail Comparator	329	0.219	180	0.8
6	A High-Speed Low-Offset Double Tail Dynamic Comparator for Low-Power Applications	11.67	0.07697	90	0.8
7	Triple-Tail Common-Mode Insensitive High-Speed Dynamic Comparator for Analog In-Memory Computing Architectures	-	0.037	65	1.1
8	Proposed Design	0.251	0.02	32	0.5

V. CONCLUSION AND FUTURE SCOPE

This paper presents detailed information about the strong-arm latch comparator. In summary, the project on the strong arm-latch comparator concludes with noteworthy achievements, boasting a minimal power dissipation of 251 nW and a commendable time delay of 0.02 ns. Table-2 represents the power dissipation of the different comparators. From the we can conclude that the strong -arm latch comparator is better for low-power applications. This outcome represents a significant leap in comparator design, effectively balancing operational speed and power efficiency. The circuit's proficiency in swiftly comparing signals with minimal energy consumption positions it as a promising candidate for applications prioritizing low power usage. These findings highlight the success of the strong arm-latch comparator design, paving the way for potential integration into diverse electronic systems. Future endeavors may explore additional optimizations and practical applications to fully exploit the advantages offered by this impressive comparator design.

The strong-arm latch comparator encompasses several avenues for improvement and innovation. Potential areas of focus include optimizing performance for speed and efficiency, reducing power consumption, adapting to advanced semiconductor technologies, enhancing noise immunity, integrating with larger systems or applications, extending capabilities to handle multi-bit comparisons, ensuring robustness and reliability under diverse conditions,

Authorized licensed use limited to: Tamkang Univ.. Downloaded on June 24,2026 at 04:34:00 UTC from IEEE Xplore. Restrictions apply.

exploring low-voltage operation, and investigating integration possibilities with machine learning or artificial intelligence. Customization for specific applications, such as medical devices or IoT sensors, is also a promising direction. Staying abreast of technological advancements and trends is crucial for ensuring the continued relevance and competitiveness of the strong-arm latch comparator in evolving scenarios.

REFERENCES

[1] Erik Backenius, Mark Vesterbacka "Comparator Design for High speed ADC's"2022.

[2] Yewale, Shubhara & Gamad, R.. (2012). Design of Low Power and High-Speed CMOS Comparator for A/D Converter Application. Wireless Engineering and Technology. 03. 10.4236/wet.2012.32015. DOI: 10.4236/wet.2012.32015

[3] Chandni Sargaiya1 , R. S. Gamad," Low Power Comparator Design for SAR-ADC" ,International Journal of VLSI System Design and Communication Systems Volume.04, IssueNo.09, September-2016, Pages: 0682-0685

[4] Khorami, A., Dastjerdi, M. B., & Ahmadi, A. F. (2016). A low-power high-speed comparator for analog to digital converters. 2016 IEEE International Symposium on Circuits and Systems (ISCAS). DOI:10.1109/iscas.2016.7538971

[5] Masaya Miyahara, Yusuke Asada, Daehwa Paik and Akira Matsuzawa, "A low-noise self-calibrating dynamic comparator for high-speed ADCs," 2008 IEEE Asian Solid-State Circuits Conference, Fukuoka, Japan, 2008, pp. 269-272. DOI: 10.1109/ASSCC.2008.4708780.

[6] B. Goll and H. Zimmermann, "A Comparator With Reduced Delay Time in 65-nm CMOS for Supply Voltages Down to 0.65 V," in IEEE Transactions on Circuits and Systems II: Express Briefs, vol. 56, no. 11, pp. 810-814, Nov. 2009.DOI: 10.1109/TCSII.2009.2030357.

[7] Chandra, K. S., & Kishore, K. H. (1943). Physical Design Implementation of High Performance CMOS Dynamic Latch Comparator. Journal of Advanced Research in Dynamical and Control Systems, ISSN No, 323-332.

[8] L. D. Teja, M. A. Reddy, B. Pravali and K. Prashanth, "Design and Analysis of CNTFET based Dynamic Comparator," 2023 4th International Conference on Electronics and Sustainable Communication Systems (ICESC), Coimbatore, India, 2023, pp. 322-327,doi:10.1109/ICESC57686.2023.10193729.

[9] R. P. Somineni and S. M. Jaweed, "Design of Low Power Multiplier using CNTFET," 2017 IEEE 7th International Advance Computing Conference (IACC), Hyderabad, India, 2017, pp. 556-559, doi: 10.1109/IACC.2017.0120.

[10] Kishore, Pinninti, P. V. Sridevi, and K. Babulu. "Design of Low Power and High Speed Braun Multiplier using Modified Gate Diffusion Input." International Journal of Computer Science and Information Security (IJCSIS) 14.11 (2016).

[11] Kishore, Pinninti, K. Babulu, and P. V. Sridevi. "Design of Low voltage, Low Power and High Speed Logic Gates Using Modified GDI.

[12] M. Ahmadi and W. Namgoong, "Comparator Power Minimization Analysis for SAR ADC Using Multiple Comparators," in IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 62, no. 10, pp. 2369-2379, Oct. 2015, doi: 10.1109/TCSI.2015.2466831.

[13] H. Zhuang, C. Tong, X. Peng and H. Tang, "Low-Power, Low-Noise Edge-Race Comparator for SAR ADCs," in IEEE Transactions on Very Large-Scale Integration (VLSI) Systems, vol. 28, no. 12, pp.2699-2707, Dec. 2020,.DOI: 10.1109/TVLSI.2020.3021680.

[14] D. S. Shylu, S. Jasmine and D. J. Moni, "A Low Power Dynamic Comparator For A 12-Bit Pipelined Successive Approximation Register (SAR) ADC," 2018 4th International Conference on Devices, Circuits and Systems (ICDCS), Coimbatore, India, 2018, pp. 339-342. DOI: 10.1109/ICDCSyst.2018.8605130.

[15] L. Ricci, L. Bertulessi and A. Bonfanti, "A low-noise high-speed comparator for a 12-bit 200-MSps SAR ADC in a 28-nm CMOS process," SMACD / PRIME 2021; International Conference on SMACD and 16th Conference on PRIME, online, 2021, pp. 1-4.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on June 24,2026 at 04:34:00 UTC from IEEE Xplore. Restrictions apply.